

Exam Date & Time: 25-Jun-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VI Semester - End Semester Makeup Examination - June -2024
Artificial Intelligence [ELE-4061]

Marks: 50

Duration: 180 mins.

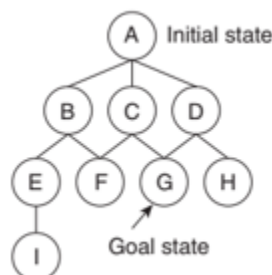
PART-A

Answer all the questions.

- 1) What is the role of artificial intelligence in today's world? (3)
 - A)
 - B) Explain PEAS representation with an example. (3)
 - C) How does an intelligent agent interact with the environment? Represent this concept with a neat block diagram. (4)
- 2) Illustrate the four conceptual components of a learning agent? Give one line description for each of them (3)
 - A)
 - B) Develop the state space representation for water jug problem. Problem definition: You have given two jugs one with capacity of four gallons and other with capacity of three gallons none is having any measuring marks on it. There is the pump that can be used to fill water in jug. How can you get exactly two gallons of water in four gallons of jug? (3)

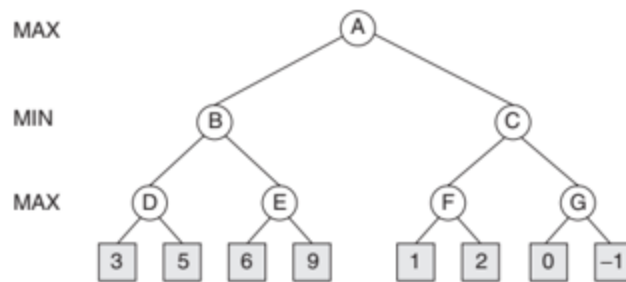


- C) Apply Breadth first search algorithm and Depth first search algorithm to discover the best path from A to G (4)



- 3) Develop the solution tree using alpha-beta pruning. Clearly show all the steps. (3)

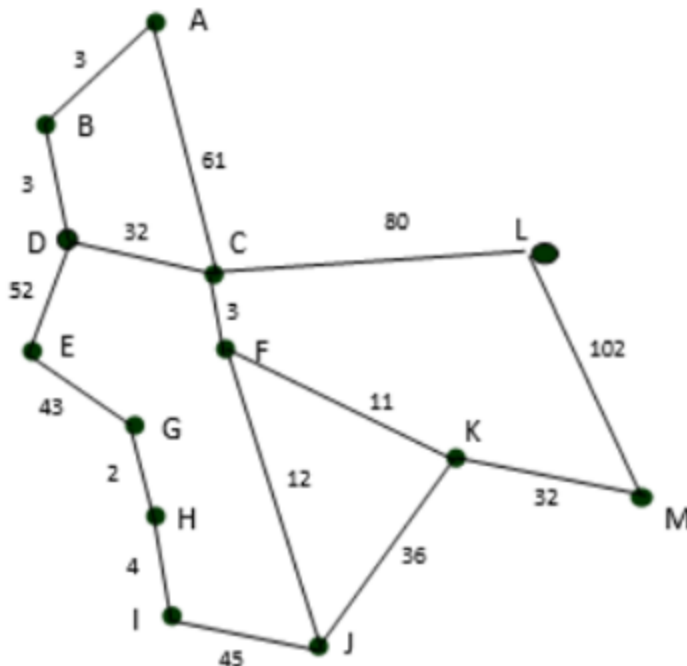
A)



B)

Apply A* algorithm to discover an optimal route from town A to town M. Heuristic values are given in the table and distance between the towns is written on edges. Show the steps clearly.

A	45
B	24
C	19
D	15
E	10
F	20
G	40
H	20
I	12
J	18
K	10
L	10
M	0



(3)

C)

Solve using Crypto arithmetic technique: SEND + MORE = MONEY

(4)

4)

What are the properties of the Wumpus World task environment?

(3)

A)

B)

Free and Bound variables are part of Quantifiers in AI. Justify the statement.

(3)

C)

Consider the following axioms.

(4)

1. John likes all kind of food
2. Apple and vegetables are food
3. Anything anyone eats and not killed is food
4. Anil eats peanuts and still alive
5. Harry eats everything that anil eats

6. Goal: John like peanuts.

Prove using Resolution.

- 5) A man is known to speak the truth 2 out of 3 times. He throws a die and reports that the number obtained is a four. Find the probability that the number obtained is actually a four. (3)

A)

- B) Consider the learning task represented by the training examples as shown below. Apply Naïve Bayes classifier, to identify the species of an entity with the following attributes.
 $X = \{\text{Color}=\text{Green}, \text{Legs}=2, \text{Height}=\text{Tall}, \text{Smelly}=\text{No}\}$

Sl.No	Color	Legs	Height	Smelly	Species
1	White	3	Short	Yes	M
2	Green	2	Tall	No	M
3	Green	3	Short	Yes	M
4	White	3	Short	Yes	M
5	Green	2	Short	No	H
6	White	2	Tall	No	H
7	White	2	Tall	No	H
8	White	2	Short	Yes	H

(3)

- C) From the given data-set, find $(x,y) = (57,170)$ whether belongs to Under or Normal Weights using KNN algorithm. Assume $K=3$

Height	Weight	Class
167	51	Underweight
182	62	Normal
176	69	Normal
173	64	Normal
172	65	Normal
174	56	Underweight
169	58	Normal
173	57	Normal
170	55	Normal
170	57	?

(4)

-----End-----